

Incident Response Report — Simulated Metasploitable 2 Compromise

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****Classification:**** Simulated — isolated home lab environment

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1. Executive Summary

In a controlled lab exercise, a Kali Linux attacker system scanned and exploited the Metasploitable 2 VM using a known FTP backdoor vulnerability. The compromise was detected through monitoring tools, and the environment was contained. Since this was an isolated lab, no external systems were impacted.

2. Scope & Environment

| Item | Detail |

|---|---|

| Attacker system | Kali Linux, IP 192.168.15.137 |

| Target system | Metasploitable 2, IP 192.168.15.129 |

| Detection tooling | Wireshark |

| Network | Isolated host-only segment, no external exposure |

3. Timeline of Events

| Time | MITRE ATT&CK Stage | Action | Evidence |

|---|---|---|---|

| 01:47 | Reconnaissance (TA0043) | Nmap scan of target | Fig. Recon-1 |

| 01:50 | Initial Access (TA0001) | Exploited vsftpd 2.3.4 via Metasploit | Fig. Exploit-1 |

| 02:09 | Execution (TA0002) | Meterpreter session established, root shell gained | Fig. Exec-1 |

| 02:15 | — | Detection alert triggered in Wireshark | Fig. Net-1 |

4. Technical Findings

4.1 Reconnaissance

Nmap revealed multiple open ports and services, including FTP (vsftpd 2.3.4), SSH, Telnet, SMTP, HTTP, MySQL, PostgreSQL, and Samba. The FTP service version was identified as vulnerable.

4.2 Vulnerabilities Exploited

- **Vulnerability:** VSFTPD 2.3.4 Backdoor (CVE-2011-2523)
- **Exploit used:** `exploit/unix/ftp/vsftpd\_234\_backdoor` (Metasploit)
- **Impact:** Remote root shell access via Meterpreter

4.3 Detection Evidence

Wireshark captured TCP traffic between attacker (192.168.15.137) and victim (192.168.15.129) on ports 4444 and 6200, confirming reverse shell activity.

5. Indicators of Compromise

Type	Value
Source IP	192.168.15.137
Target IP	192.168.15.129
Ports/services used	21/tcp (FTP), 4444/tcp (reverse shell)
Payload/signature	Meterpreter reverse TCP

6. Response Actions

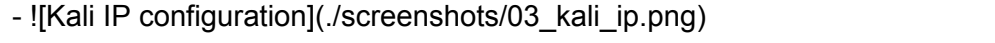
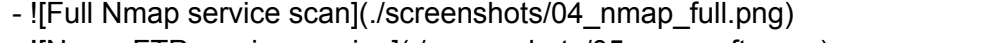
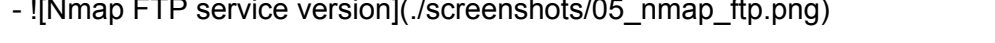
- Containment:** Isolated the Metasploitable VM from the network.
- Eradication:** Removed vulnerable vsftpd service and patched system.
- Recovery:** Restored VM to clean snapshot and confirmed no persistence.

7. Lessons Learned & Recommendations

- Outdated FTP service enabled exploitation → implement patch management.
- Lack of IDS coverage → deploy network-based intrusion detection.
- Weak segmentation → enforce stricter network isolation for test environments.
- Logging gaps → improve centralized monitoring and alerting.

8. Appendix

Figures: Reconnaissance

-  (./screenshots/01_metasploitable_ip.png)
-  (./screenshots/02_ping_target.png)
-  (./screenshots/03_kali_ip.png)
-  (./screenshots/04_nmap_full.png)
-  (./screenshots/05_nmap_ftp.png)

Figures: Initial Access & Exploitation

- ![Metasploit console startup](./screenshots/06_msfconsole.png)
- ![Search vsftpd exploit](./screenshots/07_search_vsftpd.png)
- ![Select exploit module](./screenshots/08_use_vsftpd.png)
- ![Exploit options](./screenshots/09_show_options.png)
- ![Set RHOSTS target](./screenshots/10_set_rhosts.png)
- ![Exploit configuration](./screenshots/11_exploit_config.png)
- ![Set LHOST attacker](./screenshots/12_set_lhost.png)
- ![Run exploit and gain session](./screenshots/13_run_exploit.png)

Figures: Execution & Privilege Verification

- ![Meterpreter session interaction](./screenshots/14_sessions.png)
- ![System info enumeration](./screenshots/15_sysinfo.png)
- ![Root shell access](./screenshots/16_shell_root.png)
- ![Network config from shell](./screenshots/17_ip_addr.png)
- ![Filesystem enumeration](./screenshots/18_ls_root.png)
- ![User accounts in /etc/passwd](./screenshots/19_passwd.png)

Figures: Network Detection Evidence

- ![Wireshark packet capture](./screenshots/20_wireshark.png)

This report documents a simulated attack in an isolated lab environment for training and portfolio purposes.